



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

DATA HANDLING AND VISUALIZATION

CO1 – AT2

R Programming and Data Visualization

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COURSE	DATA HANDLING AND VISUALIZATION
ASSESSMENT	CO1 – AT2

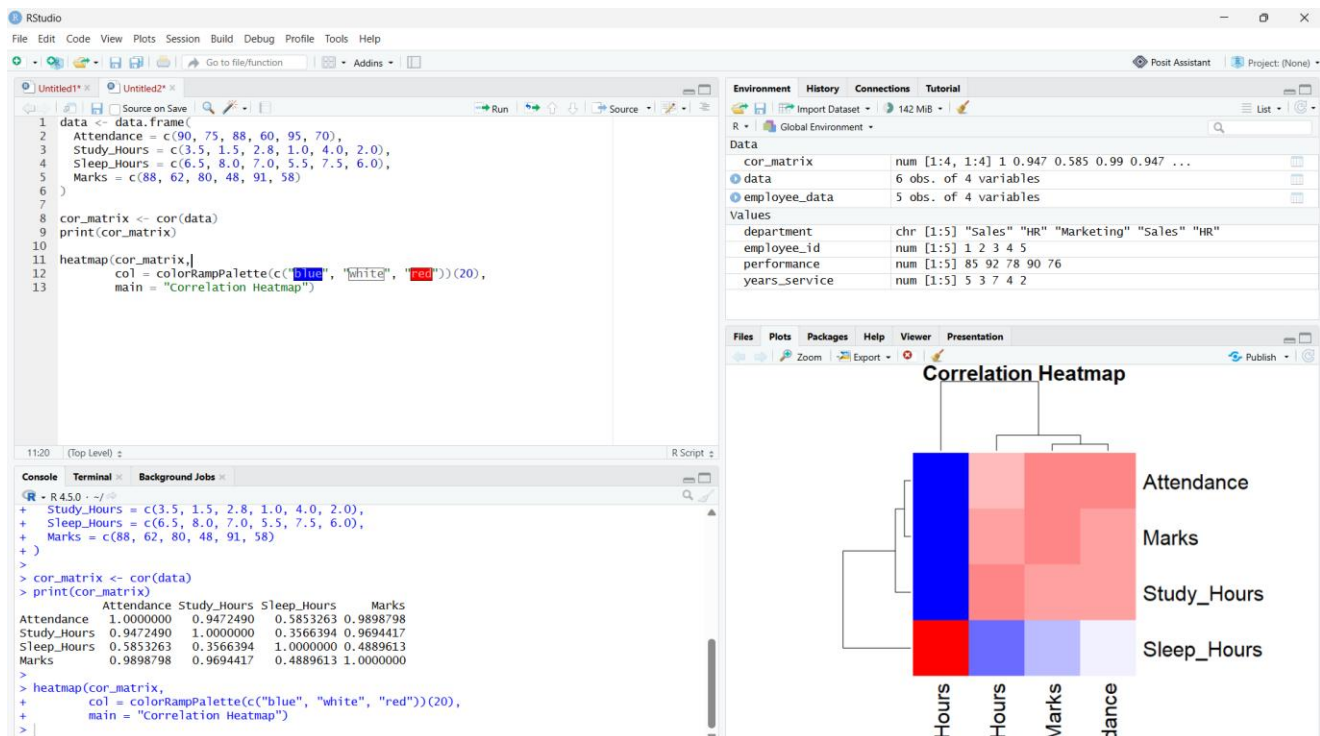
Q1. Correlation Heatmap [R]

Dataset: Student attendance, study hours per day, sleep hours per day, and marks. The purpose is to examine the relationships among these numerical variables and represent the correlation matrix as a heatmap.

Student	Attendance (%)	Study Hrs/Day	Sleep Hrs/Day	Marks (%)
S1	90	3.5	6.5	88
S2	75	1.5	8.0	62
S3	88	2.8	7.0	80
S4	60	1.0	5.5	48
S5	95	4.0	7.5	91
S6	70	2.0	6.0	58

R Program

```
data <- data.frame(  
  Attendance = c(90, 75, 88, 60, 95, 70),  
  Study_Hours = c(3.5, 1.5, 2.8, 1.0, 4.0, 2.0),  
  Sleep_Hours = c(6.5, 8.0, 7.0, 5.5, 7.5, 6.0),  
  Marks = c(88, 62, 80, 48, 91, 58)  
)  
  
cor_matrix <- cor(data)  
print(cor_matrix)  
  
heatmap(cor_matrix,  
  col = colorRampPalette(c("blue", "white", "red"))(20),  
  main = "Correlation Heatmap")
```



Interpretation

1. What is the strongest relationship?

Attendance and Marks show a very strong positive relationship in this small dataset. Students with higher attendance generally have higher marks.

2. What does the diverging color scale show?

The diverging scale uses opposite colors to distinguish negative and positive correlations, while a neutral color indicates a correlation close to zero. Darker shades indicate stronger relationships.

3. What is one limitation of correlation?

Correlation describes association, not causation. A strong correlation does not prove that changing one variable directly causes a change in the other.

4. What additional variable could improve the analysis?

Previous academic performance, assignment scores, class participation, or number of study sessions could provide additional context and help explain differences in marks.

Q2. Coordinate Systems – Cartesian and Polar [R]

Dataset: Store footfall recorded at different times of the day. The activity compares a conventional Cartesian bar chart with a polar coordinate visualization. Because time of day is cyclical, polar coordinates can provide a useful alternative representation.

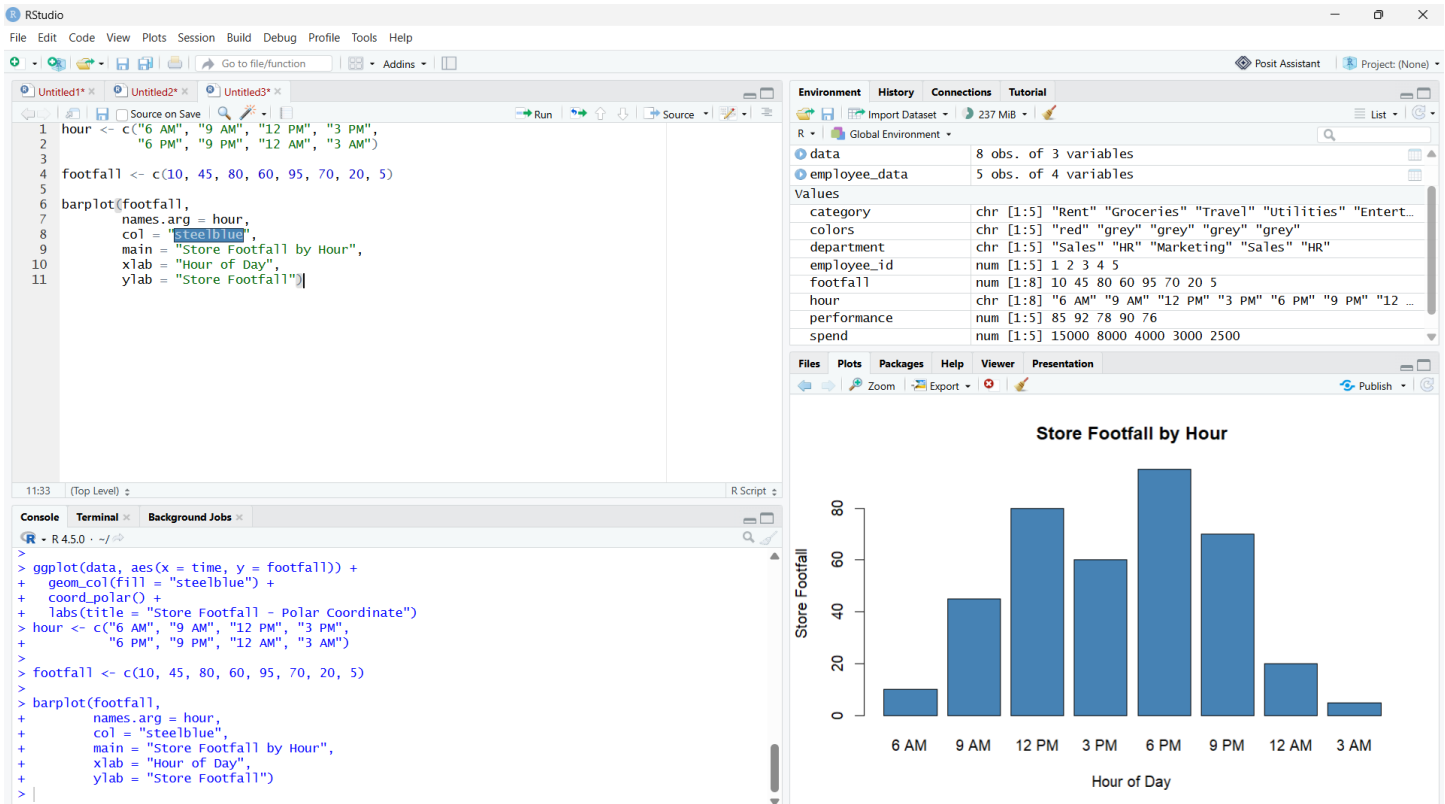
Given Data

Hour of Day	Store Footfall
6 AM	10
9 AM	45
12 PM	80
3 PM	60
6 PM	95
9 PM	70
12 AM	20
3 AM	5

Cartesian Coordinate – R Program

```
hour <- c("6 AM", "9 AM", "12 PM", "3 PM",  
          "6 PM", "9 PM", "12 AM", "3 AM")  
  
footfall <- c(10, 45, 80, 60, 95, 70, 20, 5)  
  
barplot(footfall,  
        names.arg = hour,  
        col = "steelblue",  
        main = "Store Footfall by Hour",  
        xlab = "Hour of Day",  
        ylab = "Store Footfall")
```

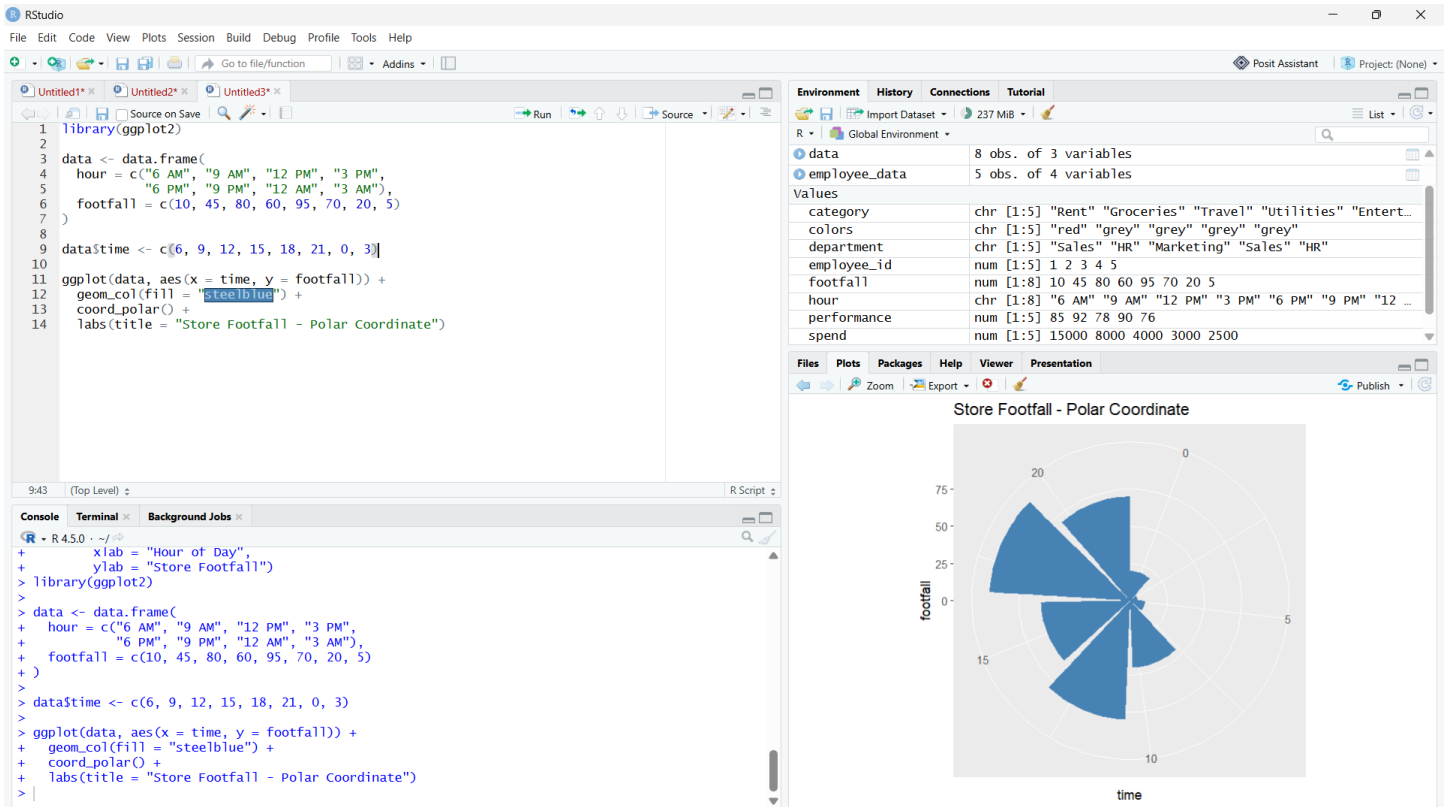
Q2A OUTPUT SCREENSHOT – CARTESIAN BAR CHART



Polar Coordinate – R Program

```
library(ggplot2)  
  
data <- data.frame(  
  hour = c("6 AM", "9 AM", "12 PM", "3 PM",  
           "6 PM", "9 PM", "12 AM", "3 AM"),  
  footfall = c(10, 45, 80, 60, 95, 70, 20, 5),  
  time = c(6, 9, 12, 15, 18, 21, 0, 3)  
)  
  
ggplot(data, aes(x = time, y = footfall)) +  
  geom_col(fill = "steelblue") +  
  coord_polar() +  
  labs(title = "Store Footfall - Polar Coordinate")
```

Q2B OUTPUT SCREENSHOT – POLAR CHART



Interpretation

1. Which chart is easier for exact comparison?

The Cartesian bar chart is generally easier for comparing individual values because the bars share a common straight baseline.

2. What is the advantage of the polar chart?

The polar chart emphasizes the 24-hour cycle. It allows the viewer to see the movement of footfall around the clock and identify periods of high and low activity.

3. Which coordinate system is suitable for cyclical time data?

Polar coordinates are useful when the cyclical nature of the variable is important. Cartesian coordinates remain useful when accurate comparison of individual values is the priority.

4. What does the data indicate?

Footfall is highest at 6 PM with 95 customers, followed by 12 PM with 80 customers. The lowest footfall occurs at 3 AM with 5 customers.

3. Color Scales for Data Representation [R]

Dataset: Employee performance scores. A sequential color scale is used because performance score is ordered from lower to higher values. The color intensity should communicate this ordering without creating unnecessary categorical distinctions.

Given Data

Employee	Performance Score
E1	62
E2	78
E3	85
E4	70
E5	91
E6	55

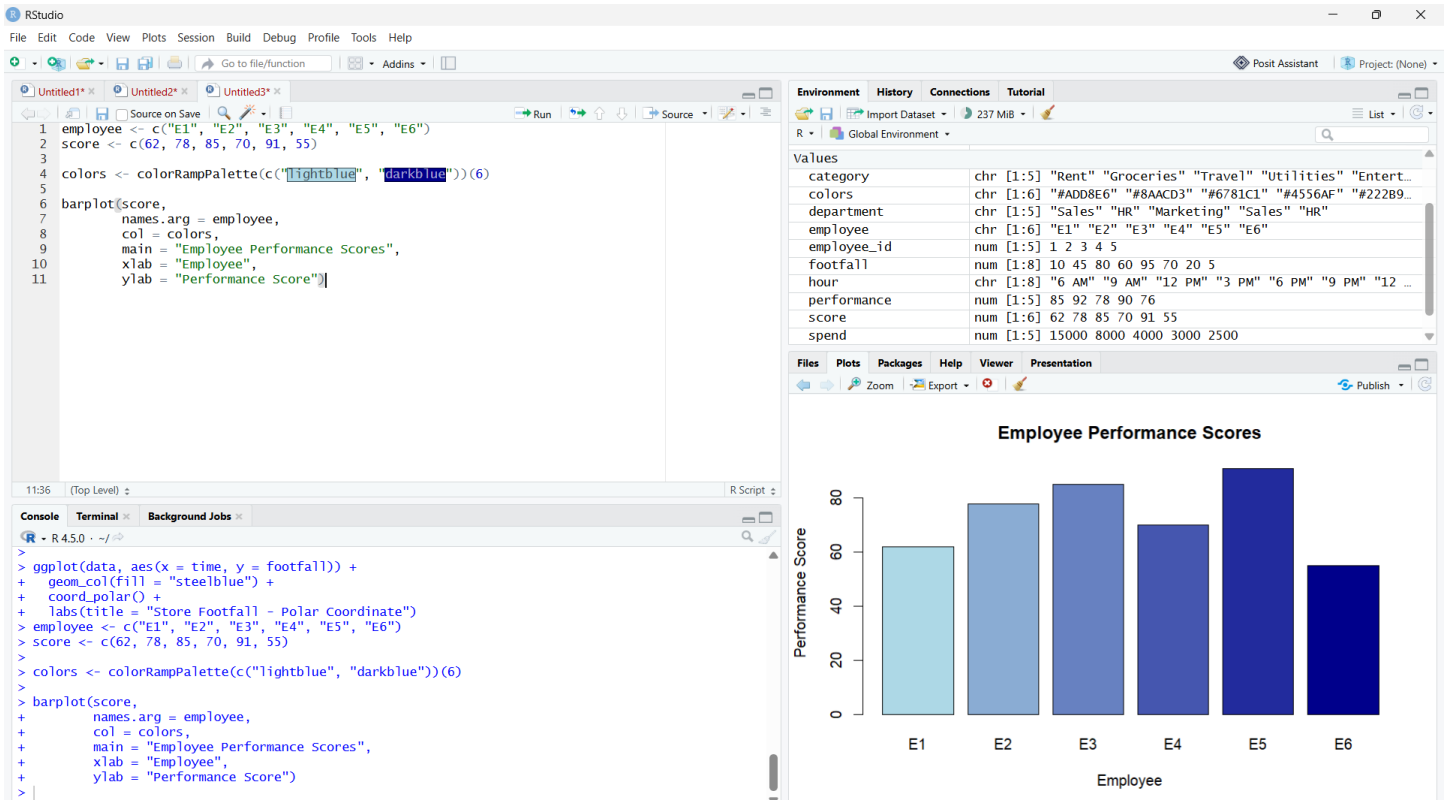
R Program

```
employee <- c("E1", "E2", "E3", "E4", "E5", "E6")
score <- c(62, 78, 85, 70, 91, 55)

colors <- colorRampPalette(c("lightblue", "darkblue"))(6)

barplot(score,
  names.arg = employee,
  col = colors,
  main = "Employee Performance Scores",
  xlab = "Employee",
  ylab = "Performance Score")
```

Q3 OUTPUT SCREENSHOT – SEQUENTIAL COLOR BAR CHART



Interpretation

1. Which employee has the highest performance?

Employee E5 has the highest performance score of 91.

2. Why is a sequential scale appropriate?

Performance scores have an ordered numerical meaning. A progression from light to dark shades can represent increasing performance in a visually intuitive way.

3. What palette can support accessibility?

A blue-based sequential palette is a reasonable choice because it avoids relying on red-versus-green differences. The viridis family is another strong option for accessibility.

4. Sequential versus categorical scales?

Sequential scales represent ordered numerical magnitude using changes in lightness or intensity. Categorical scales use distinct colors for separate groups that do not have a natural numerical order.

Q4. Integrated Visualization [R]

Dataset: Five countries with GDP, literacy rate, population, and continent. A bubble chart integrates several variables in one visualization. Position represents GDP and literacy, bubble size represents population, and color represents continent.

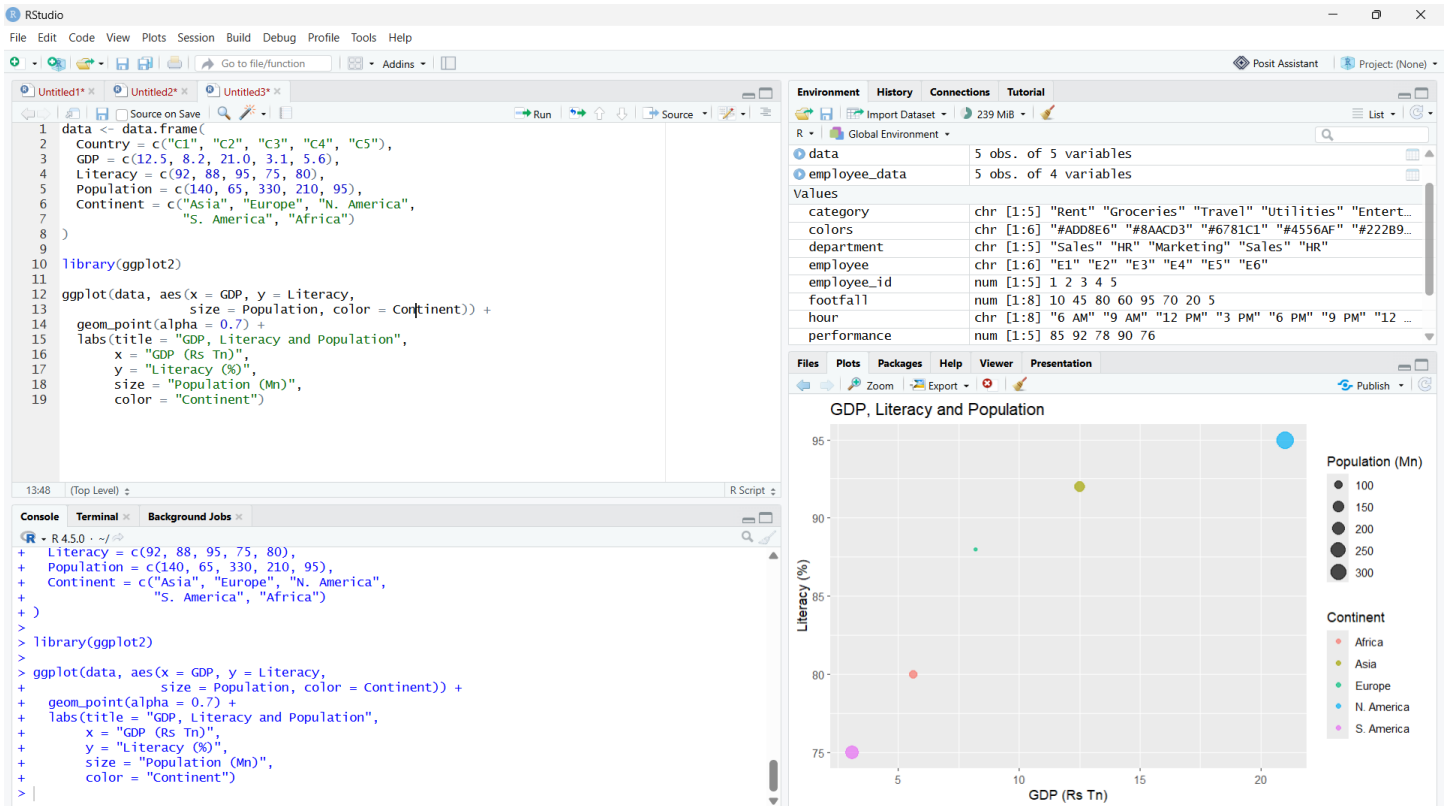
Given Data

Country	GDP (Rs Tn)	Literacy (%)	Population (Mn)	Continent
C1	12.5	92	140	Asia
C2	8.2	88	65	Europe
C3	21.0	95	330	N. America
C4	3.1	75	210	S. America
C5	5.6	80	95	Africa

R Program

```
data <- data.frame(  
  Country = c("C1", "C2", "C3", "C4", "C5"),  
  GDP = c(12.5, 8.2, 21.0, 3.1, 5.6),  
  Literacy = c(92, 88, 95, 75, 80),  
  Population = c(140, 65, 330, 210, 95),  
  Continent = c("Asia", "Europe", "N. America",  
    "S. America", "Africa")  
)  
  
library(ggplot2)  
  
ggplot(data, aes(x = GDP, y = Literacy,  
  size = Population, color = Continent)) +  
  geom_point(alpha = 0.7) +  
  labs(title = "GDP, Literacy and Population",  
    x = "GDP (Rs Tn)",  
    y = "Literacy (%)",  
    size = "Population (Mn)",  
    color = "Continent")
```

Q4 OUTPUT SCREENSHOT – INTEGRATED BUBBLE CHART



Interpretation

1. Suitable aesthetic mappings

GDP is mapped to the x-axis, Literacy Rate to the y-axis, Population to bubble size, and Continent to color.

2. What relationship can be observed?

The values suggest a generally positive association between GDP and literacy in this small dataset. For example, C3 has the highest GDP and literacy rate.

3. Continuous variables

GDP, literacy rate, and population are numerical/continuous measures in the context of this visualization.

4. Categorical variables

Country and Continent are categorical variables. Country identifies individual observations, while Continent groups countries.

5. Suggested improvement

Adding country labels and using a color-blind-friendly palette would improve readability. Transparency can also help when bubbles overlap.

Q5. Highlighting with Color [R]

Dataset: Monthly spending by expense category. The objective is to highlight the category with the highest spending so that the most important observation can be identified immediately.

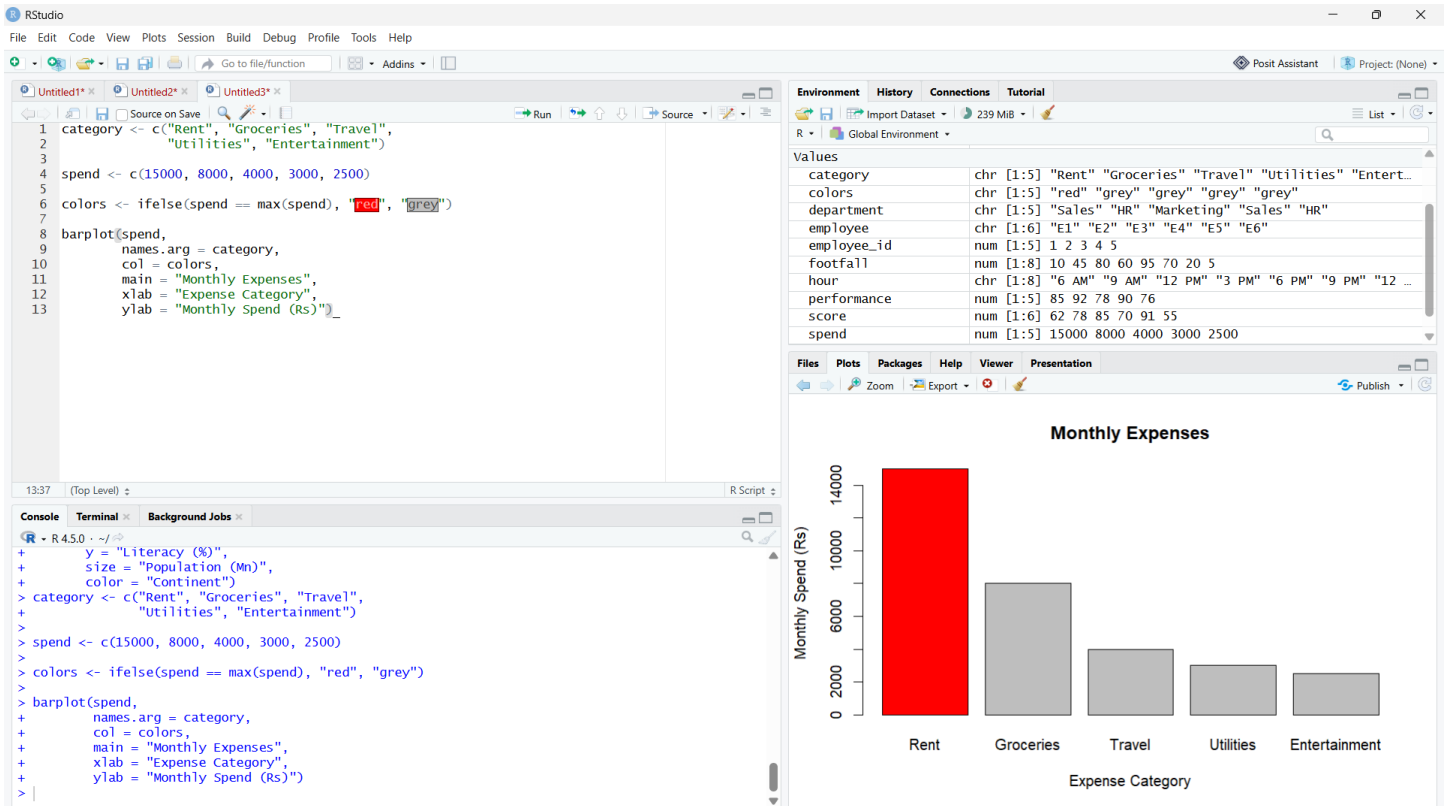
Given Data

Expense Category	Monthly Spend (Rs)
Rent	15000
Groceries	8000
Travel	4000
Utilities	3000
Entertainment	2500

R Program

```
category <- c("Rent", "Groceries", "Travel",  
             "Utilities", "Entertainment")  
  
spend <- c(15000, 8000, 4000, 3000, 2500)  
  
colors <- ifelse(spend == max(spend), "red", "grey")  
  
barplot(spend,  
        names.arg = category,  
        col = colors,  
        main = "Monthly Expenses",  
        xlab = "Expense Category",  
        ylab = "Monthly Spend (Rs)")
```

Q5 OUTPUT SCREENSHOT – HIGHLIGHTED BAR CHART



Interpretation

1. Which category has the highest spending?

Rent has the highest monthly spending at ₹15,000.

2. How does highlighting help?

Highlighting directs the viewer's attention to the most important value. In this case, the highlighted Rent bar can be identified immediately without comparing every bar in detail.

3. What happens if too many colors are used?

Excessive colors can reduce visual hierarchy, distract the viewer, and make it difficult to identify the intended focus of the chart.

4. Alternative highlighting technique

Instead of color, the highest value can be highlighted using a bold outline, a data label, an annotation, a different bar pattern, or an arrow pointing to the relevant bar.

Overall Conclusion

The five activities demonstrate important principles of data handling and visualization in R. Correlation heatmaps are useful for identifying relationships among numerical variables, while coordinate systems determine how patterns such as time cycles are displayed. Sequential color scales are appropriate for ordered numerical values, whereas categorical colors should be reserved for distinct groups.

Integrated visualizations can communicate multiple dimensions of data simultaneously by mapping variables to position, size, and color. Highlighting is useful when the objective is to draw attention to a specific observation, but it should be used carefully so that the visualization remains clear and unbiased.

A good visualization should not only be technically correct but also easy to read. Appropriate chart selection, consistent colors, meaningful labels, accessible palettes, and clear annotations all contribute to effective data communication.